Denominator Periods, Rational-Value Constraints and Achievement-Set Geometry

Mersenne subseries and Erdős Problem #257

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Abstract

Erdős Problem #257 asks whether

$$X_A = \sum_{a \in A} \frac{1}{2^a - 1}$$

is irrational for every infinite $A \subseteq \mathbb{N}_{>0}$. The problem remains open. The leading Lean-checked support-family theorem covers every infinite $A \subseteq \mathbb{N}_{>0}$ with convergent reciprocal mass, simultaneously at every integer base $b \geq 2$:

$$\sum_{a \in A} \frac{1}{a} < \infty \quad \Longrightarrow \quad \sum_{a \in A} \frac{1}{b^a - 1} \notin \mathbb{Q} \qquad (b \geq 2).$$

No pairwise-coprimality, periodicity or density hypothesis is imposed. The proof first obtains arbitrarily close positive binary returns of the shifted divisor-tail sum from exact gcd-orbit means, an LCM-prefix limit and Cesaro averaging. A uniform atom inequality transfers every such return to every larger radix, and an exact radix-b integral scaled-tail gap rules out rationality. Thus every counterexample to Problem #257 must have divergent reciprocal mass. Erdős printed this theorem in 1968 for pairwise-coprime support at every integer base $t \geq 2$, and said in the next sentence that coprimality can be removed without printing that argument. The all-base quantifier is therefore his. What is proved here is the case he withheld: the same conclusion with no arithmetic hypothesis on the support at all. No identification with Erdős's omitted argument and no theorem-priority claim are made.

For a finite nonempty support F, the reduced denominator D_F of $\sum_{n \in F}(b^n - 1)^{-1}$ satisfies

$$\mathrm{ord}_{D_F}(b) = \mathrm{lcm}(F);$$

if $\mathrm{lcm}(F) \geq 2$, then $\mathrm{lcm}(F) < D_F$. For a hypothetical rational value on an infinite support, binary long division instead produces an unbounded integral scaled-tail orbit. The orbit obeys an exact recurrence, permits only sublogarithmic zero windows in the divisor-incidence sequence, and satisfies the reciprocal-mass bounds stated below. Boolean–Möbius inversion gives an exact orbit characterisation of all rational support values, but does not make the reconstructed support infinite.

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[*] *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The achievement set of the weights $(2^n - 1)^{-1}$ is compact, perfect, totally disconnected, nowhere dense, and of Lebesgue measure 1. If the allowed exponent set has finite complement F, the restricted achievement set has measure $2^{-|F|}$; if the complement is infinite, it has measure zero. These geometric results classify sets of subsums, not the arithmetic nature of their points.

Two rational targets make the remaining difficulty explicit. Membership of $1/2$ is equivalent to infinitely many omissions in its greedy expansion; non-membership is witnessed by one finite fatal gap. Membership of $1/21$ is equivalent to cofinal crossings of the exact scaled lower separatrix, or equivalently to failure of the stated fatal aligned branch. Neither membership question is resolved. A representation of either value must have infinite support.

Strongest unconditional results

Arithmetic. Every infinite support of convergent reciprocal mass gives an irrational value at every integer base $b \geq 2$, with no coprimality, periodicity or density hypothesis. The full-support series and all nonzero eventually periodic nonnegative rational weight sequences are irrational at every integer base. Finite subsums have exact denominator order. **Geometry.** The achievement set is compact, perfect, nowhere dense, and has measure 1, with an exact measure law after restricting exponents. **Open boundary.** The target values $1/2$ and $1/21$ remain undecided, and the geometric classification does not decide the arithmetic nature of individual subsums.

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1 Introduction and main results

Problem 1.1 (Erdős #257)**.** For every infinite $A \subseteq \mathbb{N}_{>0}$, is $X_A = \sum_{a\in A} \dfrac{1}{2^a - 1}$ irrational?

See Erdős [9, p. 222], Erdős and Graham [10, p. 62], and Erdős [11, p. 105]. Bloom's current catalogue record reproduces the displayed problem and labels it open, while explicitly warning that the status is the website owner's present assessment and may omit relevant literature [12]. We therefore use the catalogue for numbering and current reported status only; the original publications and the later cited papers carry the mathematical claims.

Write $w_n = (2^n - 1)^{-1}$. Expanding each weight as a geometric series and interchanging the two nonnegative sums gives the coordinate this note works in. With the *divisor incidence* $\mathrm{sc}_A(n) = \#\{d \mid n : d \in A\}$,

$$X_A = \sum_{a\in A} \frac{1}{2^a - 1} = \sum_{n\geq 1} \frac{\mathrm{sc}_A(n)}{2^n}. \tag{1}$$

The transform is worth reading carefully, because it is where the arithmetic of the problem enters. The datum is the $0/1$ indicator of A. Equation (1) contains its divisor transform, a nonnegative integer sequence bounded by the divisor function, $\mathrm{sc}_A(n) \leq d(n)$. So #257 is not a generic question about binary digit sequences. Equation (1) is a power-series representation, not a binary expansion: its coefficients are divisor counts drawn from a single support and may exceed 1. Every theorem below is a statement about sequences of that shape.

A small instance fixes the notation. At $A = \{2, 3\}$ the incidence sequence begins

$$\mathrm{sc}_A(1), \dots, \mathrm{sc}_A(12) = 0, 1, 1, 1, 0, 2, 0, 1, 1, 1, 0, 2,$$

the value 2 occurring at the multiples of 6 and the value 0 at the integers prime to 6, and (1) reads $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$.

Several statements below hold at every integer base, so we write $X_A(b) = \sum_{a \in A}(b^a - 1)^{-1}$ for the value at an integer base $b \geq 2$, with $X_A = X_A(2)$. Base 2 is the case Problem 1.1 asks about and is meant whenever no base is named.

The reciprocal-summable support theorem. This is the result closest to the universal quantifier in Problem 1.1 that is proved here.

Theorem 1.2 (reciprocal-summable supports)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

$$\sum_{a \in A} \frac{1}{a} < \infty,$$

then $X_A(b)$ is irrational for every integer $b \geq 2$.

The exact checked declaration is

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irrational_erdosSupportSeries_of_summable_reciprocal.
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Here the Lean hypothesis is `Summable (reciprocalSupportTerm A)`; because $A \subseteq \mathbb{N}_{>0}$, this is precisely convergence of the displayed nonnegative reciprocal sum. The formal theorem has no pairwise-coprimality, periodicity or density assumption. Its all-base quantifier is part of the checked statement, not a corollary asserted only in prose.

The finite-support theorem. For a finite nonempty $F \subseteq \mathbb{N}_{>0}$ and an integer $b \geq 2$, write

$$x_F(b) = \sum_{n \in F} \frac{1}{b^n - 1} = \frac{N_F}{D_F} \qquad (\gcd(N_F, D_F) = 1,\ D_F > 0).$$

We use the standard trivial-modulus convention $\mathrm{ord}_1(b) = 1$.

Theorem 1.3 (finite-period noncollapse)**.** *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \geq 2$ be an integer, and let $D_F > 0$ be the denominator of $x_F(b)$ in lowest terms. Then D_F is coprime to b, and*

$$\mathrm{ord}_{D_F}(b) = \mathrm{lcm}\{n : n \in F\}.$$

If moreover $\mathrm{lcm}(F) \geq 2$*, then* $\mathrm{lcm}(F) < D_F$*.*

The order statement is checked, its reduced-denominator form is checked, coprimality is checked, and the growth clause is checked.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Structure. Section 2 gives the proof architecture of Theorem 1.2, including its exact integer-gap endgame and the binary-to-radix transfer. Section 3 develops the arithmetic of Theorem 1.3. Section 4 collects what rationality would force on an arbitrary infinite support, which is the part of this note that quantifies over every support rather than sampling. Section 5 gives representative supports already known to give irrational values, grouped by the argument that reaches them. Section 6 treats the squarefree support, whose values are known at every power-of-two base and which two of the arguments used here provably cannot reach at any even base. Section 7 gives the unrestricted and support-restricted topology and measure classifications, followed by the exact 1/2 and 1/21 membership criteria and unresolved branches. Section 8 states what remains.

2 Reciprocal-summable supports at every integer base

The hard step in Theorem 1.2 is not a sparse denominator estimate. It is a recurrence argument which makes the initial radix-b divisor tail a strict minimum, together with a uniform comparison which transfers arbitrarily close binary returns to every larger radix.

Let $c_A(n) = \#\{d \mid n : d \in A\}$. For $b \geq 2$, put

$$T_N^{(b)} = \sum_{n \geq 1} \frac{c_A(N+n)}{b^n}, \qquad \omega_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}.$$

Expanding each selected denominator geometrically and regrouping the nonnegative terms gives the shifted-atom identity

$$T_N^{(b)} = \sum_{d \in A} \omega_{b,d}(N). \tag{2}$$

In particular $T_0^{(b)} = X_A(b)$. Every atom is minimized at shift zero. If A is infinite and $N > 0$, choose $d \in A$ with $d > N$; then the d-atom is strictly larger at N, and hence

$$T_0^{(b)} < T_N^{(b)}. \tag{3}$$

We first work at $b = 2$. For a positive sampling step Q, the exact mean of the d-th binary atom along one orbit of positive Q-multiples is

$$M_Q(d) = \frac{\gcd(Q,d)}{d(2^{\gcd(Q,d)} - 1)}. \tag{4}$$

It satisfies $0 \le M_Q(d) \le 1/d$. Take $Q_P = \operatorname{lcm}(1, \dots, P)$. For each fixed d, eventually $d \mid Q_P$, so $M_{Q_P}(d)$ tends to $(2^d - 1)^{-1}$. The reciprocal-mass hypothesis is exactly the summable majorant needed to pass this limit through the sum over $d \in A$:

$$\sum_{d \in A} M_{Q_P}(d) \longrightarrow T_0^{(2)}. \tag{5}$$

For fixed $Q > 0$, the Cesaro means of the complete shifted sums $T^{(2)}_{(k+1)Q}$ tend to $\sum_{d \in A} M_Q(d)$. Choosing P, then a sufficiently long finite average, and finally one term no larger than that average proves

$$\forall \varepsilon > 0\ \exists N > 0 : \qquad T_0^{(2)} < T_N^{(2)} < T_0^{(2)} + \varepsilon. \tag{6}$$

The all-base step is a pointwise comparison, not a repetition of the orbit calculation. If $r = N \bmod d$, direct subtraction gives

$$\omega_{b,d}(N) - \omega_{b,d}(0) = \frac{b^r - 1}{b^d - 1}.$$

For $r = 0$ both sides below vanish. For $0 < r < d$,

$$\frac{b^r - 1}{b^d - 1} \le b^{-(d-r)} \le 2^{-(d-r)} \le 2\,\frac{2^r - 1}{2^d - 1}.$$

Consequently, uniformly in $b \ge 2$, N, and d,

$$0 \le T_N^{(b)} - T_0^{(b)} \le 2(T_N^{(2)} - T_0^{(2)}). \tag{7}$$

Applying the binary close-return statement with tolerance $\varepsilon/2$, then using (3), yields

$$\forall b \ge 2\ \forall \varepsilon > 0\ \exists N > 0 : \qquad T_0^{(b)} < T_N^{(b)} < T_0^{(b)} + \varepsilon. \tag{8}$$

The Lean-checked declarations corresponding to the atom comparison and the transferred close return are Lean and Lean.

It remains to turn a close return into an arithmetic contradiction. The atom recurrence sums exactly to

$$bT_N^{(b)} - T_{N+1}^{(b)} = c_A(N+1). \tag{9}$$

Suppose $T_0^{(b)} = X_A(b) = p/v$ with $v > 0$. Starting from $z_0 = p$, define

$$z_{N+1} = bz_N - vc_A(N+1).$$

Equation (9) gives $z_N = vT_N^{(b)} \in \mathbb{Z}$ for every N. Apply (8) with $\varepsilon = 1/v$. Then

$$0 < z_N - z_0 = v(T_N^{(b)} - T_0^{(b)}) < 1,$$

which is impossible for an integer. This proves Theorem 1.2. The exact summed recurrence and final theorem are Lean and Lean.

One immediate consequence is worth making explicit. The reciprocal sum over all powerful integers converges (use the standard representation $n = x^2y^3$, with y squarefree, and dominate by $\zeta(2)\zeta(3)$). Therefore every infinite support contained in the powerful integers, with no isolated-prime-power hypothesis at all, has irrational $X_A(b)$ at every integer base $b \ge 2$. This is a consequence of the general theorem, not a separate Comparator endpoint.

Attribution and boundary. Erdős's printed theorem on p. 222 reads: if $(n_i, n_j) = 1$ and $\sum 1/n_i < \infty$, then $\sum 1/(t^{n_i} - 1)$ is irrational for every $t \geq 2$. It is already an all-base theorem. The next sentence says that the condition $(n_i, n_j) = 1$ is superfluous and that the details are not given; p. 226 repeats that boundary. The single hypothesis Theorem 1.2 removes relative to the printed theorem is pairwise coprimality, which is exactly the hypothesis Erdős said he could remove. The base generality is his, not ours. What is added is a complete machine-checked proof of the withheld case. We do not identify this proof with Erdős's omitted proof or make a theorem-priority claim. The theorem says nothing about supports with divergent reciprocal mass. Consequently it leaves Problem 1.1 open, but it removes the entire reciprocal-summable region from the possible counterexample space.

3 Finite-support denominator periods

A finite support has a rational value. The relevant arithmetic questions are the size of its denominator and the position of the base within that denominator. Theorem 1.3 answers the second exactly and, under its stated hypothesis $\mathrm{lcm}(F) \geq 2$, gives a strict lower bound for the first. The omitted boundary is genuine: at $b = 2$ and $F = \{1\}$ the two quantities are both 1.

Six instances at $b = 2$, which also show what the hypothesis $\mathrm{lcm}(F) \geq 2$ is for:

F	$x_F(2)$	D_F	$\mathrm{lcm}(F)$	$\mathrm{ord}_{D_F}(2)$
$\{1\}$	1	1	1	1
$\{2\}$	1/3	3	2	2
$\{2, 3\}$	10/21	21	6	6
$\{1, 2, 3\}$	31/21	21	6	6
$\{2, 6\}$	22/63	63	6	6
$\{4, 6\}$	26/315	315	12	12

The last two columns agree in every row, which is the order statement. In the worked case $F = \{1, 2, 3\}$, the rational sum is $1 + 1/3 + 1/7 = 31/21$ and $2^6 \equiv 1 \pmod{21}$, whereas no smaller positive exponent gives 1. The first row is the only finite nonempty support with $\mathrm{lcm}(F) = 1$, and the only one on which $\mathrm{lcm}(F) < D_F$ fails; that is what the hypothesis $\mathrm{lcm}(F) \geq 2$ excludes.

The content is a noncollapse statement. Clearing denominators over $b^{\mathrm{lcm}(F)} - 1$ makes the period at most $\mathrm{lcm}(F)$ immediately; what is not immediate is that cancellation in the numerator cannot bring it below. The mechanism is that each selected exponent n contributes, through cyclotomic factorisation, a prime-power modulus dividing $b^n - 1$ on which b has order exactly n; no single reduction can remove all of them at once. Prime powers rather than primes is not a technicality. At $(b, n) = (2, 6)$ no prime divisor of $2^6 - 1 = 63$ has order 6 — one of the exceptional cases in Zsigmondy's theorem — while 9 does, and it is the prime power that carries the witness. The row $F = \{2, 6\}$ above is that case in full: $D_F = 63 = 9 \cdot 7$, and since 2 has order 2 modulo 3 and order 3 modulo

7, the order 6 can only come from 9. The growth clause then follows formally, since the order divides $\varphi(D_F) < D_F$, where φ is Euler's totient.

Two boundaries. This is an unconditional statement about every finite support, not a bounded table of examples, and not an implication with an open hypothesis; but it settles no infinite support, and no limit of it does. Denominator-period control is part of the classical method behind Erdős's 1948 argument [8]. Whether this exact sharp form appears in the literature has not been assessed, and no novelty is claimed.

4 Rational values and scaled tails

Section 5 below lists supports for which irrationality is known. The results here instead quantify over *every* infinite support whose value is rational, which is the quantifier in Problem 1.1. The base is 2 throughout, and each statement carries its own hypotheses.

Integral scaled tails, and unboundedness. Binary long division turns a hypothetical rational value into an integer recurrence. Fix an integer $v \geq 1$ and a sequence $f\colon \mathbb{N}_{>0} \to \mathbb{N}$ of coefficients. Call an integer sequence $u\colon \mathbb{N} \to \mathbb{Z}$ a *tempered scaled-tail sequence* for f with multiplier v if

$$u(n+1) = 2u(n) - vf(n+1) \quad \text{for every } n \geq 0, \qquad \text{and} \qquad u(n) = o(2^n).$$

The recurrence is one step of long division in base 2: double the remainder, then pay out the next coefficient. The growth condition is what pins the solution down, since two solutions of the recurrence differ by $C\,2^n$ for a constant C, and only $C = 0$ survives $u(n) = o(2^n)$. For any coefficient sequence with $f(n) \leq n$ — and $\mathrm{sc}_A(n) \leq d(n) \leq n$ qualifies, d being the divisor function — the series $\sum f(n)2^{-n}$ is rational exactly when a tempered scaled-tail sequence exists for some $v \geq 1$, and every such sequence is then the scaled tail $u(n) = v\sum_{j\geq 1} f(n+j)2^{-j}$; hence exactly one such orbit exists (checked).

The next theorem exhibits such an orbit for X_A itself, with the coefficient sequence shifted past the power of 2 in the denominator.

Theorem 4.1 (unbounded scaled-tail states)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite with $X_A = p/(2^c v)$, $v \geq 1$. Then there is $u\colon \mathbb{N} \to \mathbb{N}_{>0}$ with*

$$u(n) = v\sum_{j\geq 1} \mathrm{sc}_A(c+n+j)2^{-j}$$

satisfying the exact recurrence

$$u(n+1) + v\,\mathrm{sc}_A(c+n+1) = 2u(n), \qquad u(n) \equiv p\,2^n \pmod{v},$$

and u is unbounded.

Checked as Lean. The useful clause is exact: these scaled-tail states cannot remain bounded. In particular they cannot eventually cycle through a finite set of states.

Divisor coverage cannot have long gaps. Say that sc_A has a *zero window* of length h at N if $\mathrm{sc}_A(N+1) = \cdots = \mathrm{sc}_A(N+h) = 0$, that is, if no element of A divides any of h consecutive integers. At $A = \{2,3\}$, for instance, sc_A vanishes exactly on the integers prime to 6, so every zero window has length at most 1: one of any two consecutive integers is even.

Theorem 4.2 (sublogarithmic zero windows)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be nonempty with $X_A = p/(2^c v)$, $v \geq 1$. For every $\varepsilon > 0$ there is a constant B, depending on A, p, ε, c and v, such that every zero window of sc_A at $c + N$ has length*

$$h \leq \varepsilon \log_2(N+1) + B.$$

For fixed A, p, c, v and ε, the same B works uniformly in N and h; no common constant over supports or numerators is asserted.

Checked as Lean. Read as a constraint on counterexamples, this rules out zero windows of any fixed positive proportion of $\log_2 N$ once N is large. The proof compares an exponential lower bound forced on the scaled tail with an upper bound for divisor sums that grows more slowly than any fixed power of N.

A reciprocal-sum lower bound. Let $\mathrm{ord}_v(2)$ denote the multiplicative order of 2 modulo an odd $v > 1$, and write the reciprocal mass of A for $\sum_{a\in A} 1/a$.

Theorem 4.3 (reciprocal mass bound)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be such that $\sum_{a\in A} 1/a$ converges, and suppose $X_A = p/(2^c v)$ with $v > 1$ odd and $\gcd(p, v) = 1$. Then*

$$\sum_{a\in A} \frac{1}{a} \geq \frac{1}{\mathrm{ord}_v(2)}.$$

Checked as Lean. A finite support already supplies an instance: $A = \{2,3\}$ has $X_A = 10/21$, so $c = 0$, $v = 21$, $\gcd(p,v) = 1$ and $\mathrm{ord}_{21}(2) = 6$, and the bound reads $\frac{1}{2} + \frac{1}{3} \geq \frac{1}{6}$. The summability hypothesis is doing real work: a rational value with fixed odd denominator part v and a convergent reciprocal sum requires mass at least $1/\mathrm{ord}_v(2)$. The statement does not apply when the reciprocal sum diverges, and it gives no positive lower bound uniform in v; it constrains the pair (support, denominator) jointly.

The critical dyadic case has a different endpoint.

Theorem 4.4 (dyadic reciprocal-mass endpoint)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite and suppose $X_A = p/2^c$ for some $p \in \mathbb{Z}$ and $c \in \mathbb{N}$. Then the reciprocal support terms are not summable, or*

$$\sum_{a\in A} \frac{1}{a} > 1.$$

Checked as Lean. The two alternatives are an exact necessary consequence of a dyadic rational value. They do not contradict rationality by themselves: a contradiction requires separate proofs that the reciprocal terms are summable and that the mass is at most 1. The formal nonsummability alternative also does not, by itself, assert a particular asymptotic law for the partial sums.

A Boolean–Möbius characterisation. The preceding theorems give necessary conditions. The next statement is an equivalence, and is the sharpest description of rational-valued supports the development has. It removes the support from the description altogether, which is possible because the support can always be read back off the incidence sequence: with μ the Möbius function, $*$ Dirichlet convolution $(g * h)(n) = \sum_{d|n} g(d)h(n/d)$ and $\mathbf{1}_A$ the indicator of A, the identity $\mathrm{sc}_A = \mathbf{1}_A * 1$ inverts to $\mu * \mathrm{sc}_A = \mathbf{1}_A$.

For $p \in \mathbb{Z}$, $q \geq 1$ and $U : \mathbb{N} \to \mathbb{Z}$, set

$$Q_U(0) = 0, \qquad Q_U(n) = \frac{2U(n-1) - U(n)}{q} \quad (n \geq 1).$$

Call U an *admissible Boolean–Möbius scaled-tail sequence* for (p, q) if

$$\begin{gathered} U(0) = p, \quad U(N) > 0, \quad U(N) \leq q(2\sqrt{N} + 4) \quad (N \geq 0), \\ q \mid 2U(N) - U(N+1) \quad (N \geq 0), \qquad (\mu * Q_U)(n) \in \{0, 1\} \quad (n \geq 1). \end{gathered} \tag{10}$$

The divisibility condition makes Q_U integral; the last condition says that Möbius inversion of the quotient is the indicator of a set.

Theorem 4.5 (Boolean–Möbius scaled-tail correspondence)**.** *Let $p \in \mathbb{Z}$ and $q \geq 1$. There exists a support A with $0 \notin A$, with some positive element, and with $X_A = p/q$, if and only if there exists an admissible Boolean–Möbius scaled-tail sequence U for (p, q). In that case the support is recovered as*

$$A = \{n : (\mu * Q_U)(n) = 1\},$$

with Q_U as in (10).

Checked as Lean. The finite example $A = \{2, 3\}$ makes the definition concrete. Here $p/q = 10/21$, and

$$Q_U(1), \dots, Q_U(6) = 0, 1, 1, 1, 0, 2, \qquad U(0), \dots, U(6) = 10, 20, 19, 17, 13, 26, 10.$$

Möbius inversion gives $(\mu * Q_U)(1), \dots, (\mu * Q_U)(6) = 0, 1, 1, 0, 0, 0$, the indicator of $\{2, 3\}$ through that range. This finite calculation illustrates the correspondence; the admissible scaled-tail sequence itself is an infinite object, and the finite support is not a counterexample to Problem 1.1.

One boundary on the equivalence. The support it produces is not required to be infinite, and a finite support supplies an admissible scaled-tail sequence for its own value, so existence of such a sequence is not by itself a counterexample to Problem 1.1. What the equivalence changes is the search space, not the difficulty.

Combined constraints on a rational counterexample. Taken together: a counterexample to Problem 1.1 would have an unbounded scaled-tail sequence obeying an exact linear recurrence and sublogarithmic divisor-coverage gaps. If its reciprocal sum converged and its reduced odd denominator part were greater than 1, it would also satisfy Theorem 4.3; at a dyadic rational value it would instead satisfy Theorem 4.4. Theorem 4.5 reconstructs every rational value, but by itself does not force the reconstructed support to be infinite. These qualifications matter: the statements have different hypotheses, and no jointly contradictory combination has been proved.

5 Representative known irrational supports

The following table is representative, not exhaustive. It groups the displayed supports by six mechanisms rather than suggesting that its rows classify all known cases.
In the sentence immediately following the p. 222 theorem, Erdős says that pairwise coprimality can be removed by a more complicated argument, but he does not give that argument; p. 226 repeats that boundary. His printed theorem already quantifies over every integer base $t \geq 2$, so the base generality of the first row is his. The first row records a complete Lean-checked proof of the conclusion he stated and withheld, namely the same theorem with pairwise coprimality dropped. The sparse-support row separately records his printed pairwise-coprime theorem and its formalisation. Removing coprimality is the contribution, and it carries no priority claim.

Under the additional hypothesis that the pairwise-coprime exponents are odd, Duverney, Suzuki and Tachiya strengthen this conclusion: Corollary 1 gives linear independence of (1) and three associated plus/minus reciprocal sums at every integer base of absolute value greater than one [2, Corollary 1]. This is a stronger conclusion on a narrower support class; it does not remove the hypothesis from the row above.
Five remarks on the table: one on containment between the mechanisms, three on the literature rows, and one on what "checked here" means.

For the full-support row, the weighted certificate theorem is the exact checked statement; the support-series formulation displayed in the table is its later corollary.

The mechanisms are not nested, and one non-containment is proved. The base-b full-support theorem is exactly the $A = \mathbb{N}_{>0}$ statement; the multiples rows genuinely specialise it after a base change, but the sparse rows do not. The distinction is intrinsic: the denominator-gap criterion behind the factorial and power-of-two rows *provably cannot* reach the full support: the prefix lcm grows too slowly for its hypothesis to hold (checked). Conversely the analytic method of [5] reaches the primes, which are neither eventually periodic nor pairwise coprime with summable reciprocals. No list contains another, and their union does not exhaust the infinite supports.

One theorem of Duverney and Tachiya generates several rows. Their refinement of the Chowla–Erdős method [4, Cor. 1.2, author-preprint p. 4; proof pp. 9–11] proves, for a pairwise coprime sequence E of polynomial growth and the set $F_s(E)$ of products of its members with exponents below s, that 1 and the values $\sum_{n \in F_s} (q^{jn^i} - 1)^{-1}$ are linearly independent over $\mathbb{Q}$ whenever $|q|^L \leq s$, with $L = \operatorname{lcm}(1, \ldots, \ell)$ and ℓ bounding the exponent i. This is a row-generating theorem, not an isolated example.

With E the primes, $s = \infty$ returns the full support at every base, while $s = 2$ returns the squarefree support. For $s = 2$, the constraint forces $\ell = 1$ and $|q| \leq 2$, but the free exponent j gives every base $b = 2^j$. Thus this theorem excludes bases that are not powers of 2, rather than all bases $b \geq 3$; it also excludes higher monomial degrees $i \geq 2$. Their conclusion is stronger than irrationality, being linear independence of the whole finite family across the chosen exponents j.

Support A	Bases	Mechanism and authority
LCM-prefix orbit means and an integral tail gap		
arbitrary infinite A with $\sum_{a \in A} a^{-1} < \infty$	every $b \geq 2$	Theorem 1.2; checked here, with exact boundary in Section 2
Classical full support, and base change into it		
all of $\mathbb{N}_{>0}$	every $b \geq 2$	Erdős 1948 [8]; formalised here
multiples of a fixed d	every $b \geq 2$	dilation: the multiples series at base b *is* the full-support series at base b^d (checked)
Periodicity of the coefficient sequence		
eventually periodic	every $b \geq 2$	checked here; Luca–Tachiya prove the nonnegative purely-periodic case [6, Theorem 1, p. 139; proof pp. 149–150]; finite rational prefixes give the infinite eventual case
a residue class; the odd numbers	every $b \geq 2$	special cases of the row above (residue class, odd); Luca–Tachiya's Example 2 strengthens the odd row to joint linear independence of every finite divisor-convolution ladder, also for negative integer bases with absolute value greater than one [6, Example 2, p. 140]
Denominator gaps along a sparse support		
factorials $\{n!\}$	every $b \geq 2$	checked here
powers of two $\{2^k\}$	every $b \geq 2$	checked here
pairwise coprime, $\sum_{a \in A} a^{-1} < \infty$	every $b \geq 2$	Erdős [9], theorem on p. 222; formalised here
Multiplicative closure: Chowla–Erdős, refined		
$F_s(E)$ and the monomial images $\{jn^i : n \in F_s(E)\}$	$b = q^j$ with $\lvert q \rvert^L \leq s$, $L = \operatorname{lcm}(1, \ldots, \ell)$	Duverney–Tachiya [4, Cor. 1.2, author-preprint p. 4]; linear independence, not only irrationality; not formalised here
s-free positive integers, with or without 1	$b = q^j$, $2 \leq \lvert q \rvert \leq s$, $j \geq 1$	*ibid.*, with E the primes and $\ell = i = 1$; deleting 1 changes the value by $1/(b-1) \in \mathbb{Q}$
squarefree	every $b = 2^j$, $j \geq 1$	Duverney–Tachiya [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4]; joint linear independence for every finite set of such bases; not formalised here
coprime to a fixed N; sums of two squares	every $b \geq 2$	*ibid.*, Examples 1.3 and 1.2; not formalised here
Analytic		
primes	$b = 2$ proved	Tao–Teräväinen [5], Thm. 1.3, p. 4; proof pp. 44–56; not formalised here
primes, $b \geq 3$; prime powers	—	*ibid.*, asserted as modifications with details omitted in the cited version
arbitrary infinite A	—	**Open** (Problem 1.1)

The classical full-support value now has a digit-level refinement. At base 2, Campbell writes the Erdős–Borwein constant in the equivalent forms

$$E = \sum_{n\geq 1} \frac{1}{2^n - 1} = \sum_{n\geq 1} \frac{d(n)}{2^n}$$

and proves that the binary block 11 occurs infinitely often in its base-2 expansion [3, Theorem 1, p. 12; proof pp. 12–24]. This adds genuine digit-distribution information to the full-support row, but it neither proves normality nor addresses an arbitrary infinite support A, so it does not change the open status of Problem 1.1.

The two analytic rows do not have the same standing. Theorem 1.3 on p. 4, proved in Section 5 on pp. 44–56, of [5] proves the prime-support case at base 2, the series there being $\sum_{n\geq 1} \omega(n)/2^n$. The extension to every integer base, and the prime-power support — which those authors themselves identify with Problem 1.1 — are asserted by remark, with the modifications explicitly left to the reader. The table keeps the three apart, and only the first is proved.

Formalisation is not priority. Rows marked "checked here" are Lean statements accepted by the pinned kernel. For the full support that is a formalisation of Erdős; Luca and Tachiya's RIMS paper already proves the nonnegative purely-periodic case; a finite rational-prefix correction gives the eventual-periodic extension used here [6, Theorem 1, p. 139; proof pp. 149–150], while Theorem A restates the broader signed purely-periodic theorem without reproducing its earlier proof. No priority is claimed anywhere in this table.

Remark (signed periodic weights: a gap in the method, not in the mathematics). Dropping nonnegativity weakens the conclusion available, and it is worth being exact about where. For $u\colon \mathbb{N}_{>0} \to \mathbb{Z}$ periodic and $b \geq 2$, what is checked here is a *dichotomy*: the series $\sum_{n\geq 1} u(n)/(b^n - 1)$ is irrational, *or* some power of b times its value is an integer (checked). For a nonnegative support indicator the terminating alternative is excluded by the rows above; for mixed signs this argument does not exclude it.

The terminating alternative never in fact occurs. Theorem A in Luca and Tachiya's 2017 RIMS paper explicitly restates their earlier Theorem 1.1 in the exact form needed here: $\sum_{n\geq 1} a_n/(q^n - 1)$ is irrational for every integer q with $|q| > 1$ and every nonzero purely periodic integer sequence (a_n) [6, Theorem A, p. 139]. Thus the disjunction above is a consequence of the chosen argument, not a feature of the problem: the second branch is empty, and their author restatement is what shows it. The RIMS paper does not reproduce the earlier proof or verify the broader eventual-rational formulation, so neither is claimed here. The local statement retains only the interest of being an independently checked argument with a visible boundary. Section 6 records a second, sharper instance of the same phenomenon.

6 Squarefree support and a coordinate-dependent obstruction

Let $A_{\mathrm{sf}} = \{d \geq 2 : d \text{ squarefree}\}$: a support of density $6/\pi^2$, far denser than the primes, and not periodic. Its values at all power-of-two bases are jointly linearly independent with 1, by a theorem of Duverney and Tachiya recalled below. This section establishes that two block-certificate arguments used here cannot reach that support at any even base, and that the reason lies in a normalisation rather than in the value.

Theorem 6.1 (squarefree divisor incidence)**.** *For every $n \geq 1$ the number of squarefree divisors of n is $2^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors, and hence*

$$\mathrm{sc}_{A_{\mathrm{sf}}}(n) = 2^{\omega(n)} - 1.$$

In particular $\mathrm{sc}_{A_{\mathrm{sf}}}(n)$ is odd *for every $n \geq 2$.*

The count is checked, the incidence formula is checked, and the parity conclusion is checked. The proof is the bijection between squarefree divisors of n and subsets of its prime factors (Lean), so the count is $2^{\omega(n)}$; removing $d = 1$ leaves an odd number whenever $\omega(n) \geq 1$. At $n = 12 = 2^2 \cdot 3$ the squarefree divisors are $1, 2, 3, 6$, so $\mathrm{sc}_{A_{\mathrm{sf}}}(12) = 3$; at $n = 30$ they are the eight divisors of 30 and $\mathrm{sc}_{A_{\mathrm{sf}}}(30) = 7$.

6.1 The values are known at every power-of-two base

Duverney and Tachiya's Corollary 1.2, applied with E the primes, $s = 2$ and $\ell = 1$, gives $L = \mathrm{lcm}(1) = 1$ and the admissibility condition $|q| \leq 2$. The exponent j in their displayed family remains arbitrary. Thus, for every $h \geq 1$, their Example 1.1 says that the numbers

$$1, \qquad \sum_{n \geq 1} \frac{|\mu(n)|}{2^{jn} - 1} \quad (1 \leq j \leq h)$$

are linearly independent over $\mathbb{Q}$ [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4]. Since $|\mu|$ is the indicator of the squarefree integers including $n = 1$, and A_{sf} omits only $n = 1$, whose weight at base 2^j is $(2^j - 1)^{-1} \in \mathbb{Q}$,

$$X_{A_{\mathrm{sf}}}(2^j) = \sum_{n \geq 1} \frac{|\mu(n)|}{2^{jn} - 1} - \frac{1}{2^j - 1},$$

and rational translation preserves the joint linear independence with 1.

Corollary 6.2 (joint power-of-two-base theorem)**.** *For every $h \geq 1$, the $h + 1$ numbers*

$$1, \quad X_{A_{\mathrm{sf}}}(2), \quad X_{A_{\mathrm{sf}}}(4), \quad \ldots, \quad X_{A_{\mathrm{sf}}}(2^h)$$

are linearly independent over $\mathbb{Q}$. In particular, every $X_{A_{\mathrm{sf}}}(2^j)$, $j \geq 1$, is irrational.

Two boundaries on that. It is a citation, not a formalisation: nothing in this development proves it. The admissibility condition $|q|^L \leq s$ at $s = 2$ allows only $|q| = 2$, so this citation covers the bases 2^j and does not cover bases that are not powers of 2; this note proves no result about those remaining values.

6.2 Two block-certificate hypotheses have no instance

Fix $b \geq 2$ and a nonnegative coefficient sequence f. The two hypotheses used below are best printed rather than named. For every precision $q \geq 1$, they ask for $N, K, L, C \in \mathbb{N}$ with $K \leq L$ satisfying the common conditions

$$\sum_{r=K+1}^{L} f(N+r)b^{L-r} \leq C, \qquad \exists t \in \mathbb{N} : f(N+L+1+t) > 0, \qquad q(C+N+L+2) < b^L, \tag{11}$$

together with one of the two first-block conditions

$$\begin{aligned} &\text{digitwise:} \quad b^r \mid f(N+r) \quad (1 \leq r \leq K), \\ &\text{weighted:} \quad b^K \mid \sum_{r=1}^{K} f(N+r)b^{K-r}. \end{aligned} \tag{12}$$

Thus the data N, K, L, C may depend on q. The parity of Theorem 6.1 refutes both alternatives for the squarefree incidence sequence at every even base.

Corollary 6.3 (neither divisibility-first hypothesis has an instance)**.** *For the squarefree support A_{sf} and every even base $b \geq 2$, neither the digitwise nor the carry-aware block-certificate hypothesis holds. Both fail already at precision $q = b^2$.*

Proof. The hypothesis quantifies over every precision, so it is enough to exhibit one precision at which no admissible data exist; we take $q = b^2$ and split on whether the first block is empty.

Both hypotheses ask, for every precision q, for N, $K \leq L$ and C with a first-block condition on $\mathrm{sc}_{A_{\mathrm{sf}}}(N+1), \dots, \mathrm{sc}_{A_{\mathrm{sf}}}(N+K)$, a middle bound $\sum_{r=K+1}^{L} \mathrm{sc}_{A_{\mathrm{sf}}}(N+r)b^{L-r} \leq C$, a nonzero coefficient beyond $N+L$, and $q(C+N+L+2) < b^L$. Take $q = b^2$ and write $f = \mathrm{sc}_{A_{\mathrm{sf}}}$.

Suppose $K \geq 1$. The digitwise condition requires $b^r \mid f(N+r)$ for $1 \leq r \leq K$. If $N + K \geq 2$, its instance at $r = K$ makes the even number b divide the odd number $f(N+K)$. The carry-aware condition requires $b^K \mid \sum_{r=1}^{K} f(N+r)b^{K-r}$; reducing modulo b kills every term but $r = K$, so the same contradiction follows when $N + K \geq 2$.

The only remaining case with $K \geq 1$ is $N = 0, K = 1$. If $L = 1$, then $q(C+N+L+2) \geq 3b^2 > b$; if $L \geq 2$, the $r = 2$ term in the middle sum is $f(2)b^{L-2} = b^{L-2}$, so $qC \geq b^L$. Both contradict the strict size inequality.

Suppose $K = 0$, where both first-block conditions are vacuous. If $L = 0$ the middle sum is empty and $q(C+N+2) < b^0 = 1$ is impossible. If $L \geq 1$ the $r = 1$ term gives $C \geq b^{L-1}$ whenever $N \geq 1$, and again the size inequality fails. If $N = 0$ and $L = 1$, its left side is at least $3b^2 > b$; if $N = 0$ and $L \geq 2$, the $r = 2$ term gives $C \geq b^{L-2}$ and hence $qC \geq b^L$. These exhaust the cases. $\square$

The two formal nonexistence statements are checked directly as the carry-aware no-go and the digitwise no-go. The displayed proof is retained to expose the empty- and one-position-block boundary cases rather than leaving the scope hidden behind the interfaces.

6.3 The obstruction is a normalisation, not the value

Corollary 6.3 is a statement about two arguments failing on a value that Corollary 6.2 shows to be irrational. The question it raises is therefore not whether the value can be reached, but what the failure is a property *of*. It is a property of where the support starts.

Adjoin 1 to the support and write $A_{\mathrm{sf}}^{+} = A_{\mathrm{sf}} \cup \{1\}$, the full squarefree support. Then

$$X_{A_{\mathrm{sf}}^{+}}(b) - X_{A_{\mathrm{sf}}}(b) = \frac{1}{b-1} \in \mathbb{Q},$$

so the two supports pose the same irrationality question at every base, while the divisor incidence changes from $2^{\omega(n)} - 1$ to $2^{\omega(n)}$ — from odd to even at every $n \geq 2$. The parity obstruction of Corollary 6.3 evaporates under a shift that provably cannot change the answer. The shifted coefficient identity and the exact equivalence of the two irrationality questions are checked as $2^{\omega(n)}$ incidence and the shift iff.

More is true: the shifted first-block condition at base 2 asks for $2^r \mid 2^{\omega(N+r)}$, that is $\omega(N+r) \geq r$ for $1 \leq r \leq K$, and a Chinese-remainder construction reserving r fresh primes for each shift r supplies such an N for every K. This is checked both as the arithmetic block theorem for ω and in the corresponding formal statement for the shifted incidence.

It does *not* follow that either argument certifies the shifted support: the opening block is one of the conditions listed above, and the middle bound and the arithmetic inequality are untouched by this observation. What does follow is a methodological point: an obstruction stated against a coefficient sequence can be an artefact of the normalisation chosen for that sequence. Before a no-go result is reported as a property of a problem, the coordinate it is stated in should be varied by a transformation the problem is known to be invariant under. Here the transformation is adding one rational number, and it removes the obstruction entirely.

The same invariance holds for every finite change, not only this one. If $A \triangle B$ is finite, choose M above all of its elements. The two support series then have the same tail beyond M, while each omitted prefix is rational. The two directions of this argument are exactly the checked prefix lemmas Lean and Lean. Thus $X_A(b)$ is irrational if and only if $X_B(b)$ is irrational for every integer $b \geq 2$. The present shift is the smallest instance of a general checked finite-change principle.

This is the second time in this note that a boundary turns out to belong to the method rather than to the mathematics. In Remark 5 the terminating alternative of the signed periodic dichotomy is empty, and a theorem of Luca and Tachiya is what shows it; here a parity obstruction survives only until the support is shifted by one element. In each case the correction came from outside the development — once from the literature, once from asking what the statement was invariant under. A formalised no-go result carries exactly the authority of its hypotheses, and its hypotheses include the coordinates it was written in.

7 Achievement-set geometry and the value $1/2$

Problem 1.1 asks whether every value obtainable from infinitely many of the weights $w_n = (2^n - 1)^{-1}$ is irrational. The set of all values obtainable, from finite and infinite selections alike, is the achievement set $\mathcal{A} = \{\sum_{n\geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0,1\}\}$, and this section is about its geometry.

Theorem 7.1 (geometry of $\mathcal{A}$)**.** *$\mathcal{A}$ is compact, perfect, totally disconnected and nowhere dense, and has Lebesgue measure* 1*.*

Checked as compact, perfect, totally disconnected, nowhere dense, and of measure one. So $\mathcal{A}$ is a fat Cantor set: strict tail domination $\sum_{\ell>n} w_\ell < w_n$ opens a gap at every level, while the total measure is not lost. The classical antecedent is Kakeya's two-page paper: he proves perfectness for the subsum set of every absolutely convergent real or complex series and gives the real interval criterion; under the present strict tail inequality at every index, his result also gives nowhere density [1, p. 251]. The strict inequality, the resulting distinctness of subsums over distinct supports, and the Cantor conclusion for every integer base are also recorded in Remark 4.1 (p. 13) of Kovač and Tao [7]; no novelty is claimed for any of these facts. That remark makes no metric assertion.

Strict tail domination does not determine the measure: the weights 3^{-n} satisfy $\sum_{\ell>n} 3^{-\ell} = 3^{-n}/2 < 3^{-n}$ and produce a null achievement set, the base-3 digits-in-$\{0,1\}$ Cantor set. So the measure-one clause is added here rather than formalised from there, and it is the arithmetic of the Mersenne weights that supplies it.

With $T_n = \sum_{k>n} w_k$, the standard level-n convex-hull cover consists of 2^n disjoint intervals of length T_n, its nested intersection is $\mathcal{A}$, and

$$2^n T_n = \sum_{j\geq 1} \frac{2^n}{2^{n+j}-1} \longrightarrow \sum_{j\geq 1} 2^{-j} = 1,$$

dominated by 2^{1-j}. Continuity of measure from above therefore gives $\lambda(\mathcal{A}) = 1$.

Membership is characterised level by level, by a greedy expansion. Run through $n = 1, 2, \ldots$ carrying a remainder, initially the target x, and at level n subtract w_n from the remainder if w_n does not exceed it, leaving the remainder unchanged otherwise; call a level at which nothing is subtracted *skipped*. Say that the expansion *survives* level n if the remainder after that level is at most the remaining mass T_n. Then $x \in \mathcal{A}$ if and only if $x \geq 0$ and the expansion survives every level (Lean).

Non-membership of a nonnegative target is therefore visible at a single level, and for a rational target strict failure can be certified effectively: rational upper bounds for the remaining tail converge to T_n, so once one lies below the rational residual the fatal inequality is proved. The finite example below uses an exact rational upper bound. For $x = 3/4$, the greedy algorithm skips $w_1 = 1$ and leaves residual $3/4$, while the exact zero-lookahead upper bound for the remaining tail is $2w_2 = 2/3 < 3/4$; hence $3/4 \notin \mathcal{A}$ (Lean).

The selector-to-subsum map sending a 0/1 string (ε_n) to $\sum_n \varepsilon_n w_n$, which the strict tail inequality makes injective, remains injective after restriction to any subfamily. That matters because Problem 1.1 quantifies over every infinite support rather than over $\mathbb{N}_{>0}$. For a set J of future offsets write $T_J(n) = \sum_{k\in J} w_{n+k+1}$ for the tail restricted to J (definition, summable at Lean).

Then $T_J(n) < w_n$ for every J and every $n \geq 1$ (checked), since deleting weights only shrinks a tail that already sits below w_n at the full support. Consequently the digit map is injective on strings supported in J (checked), stated on the subtype of digit strings vanishing off J and evaluated by the restricted digit map, so the statement is about the subseries itself and not a projection of the full one. Thus uniqueness of the selector survives passage to an arbitrary subfamily; this is an injectivity statement, not a statement about binary digits of the value.

That is a constraint on the shape of an argument, not on the supports: it decides no value, and it is weaker than any statement in Section 4.

7.1 Geometry after restricting the allowed exponents

The injectivity statement has a geometric completion for every set $J \subseteq \mathbb{N}$ of allowed digit positions. Let

$$\mathcal{A}_J = \left\{ \sum_{k \in J} \varepsilon_k w_{k+1} : \varepsilon_k \in \{0,1\} \right\}.$$

Formally, the allowed strings form the supported digit set, which is closed. Its range is the restricted achievement set; the range and image descriptions agree by the image theorem. Consequently $\mathcal{A}_J$ is compact and closed.

It lies inside $\mathcal{A}$ by support restriction and is therefore nowhere dense. If J is infinite, the digit space is preperfect; injectivity transfers that property to $\mathcal{A}_J$, so the closed set is perfect. Thus every infinite allowed support gives a compact perfect nowhere-dense set with a unique selector for each point.

The metric classification is exact. The summability lemma controls digit terms, and changing one coordinate changes the value by precisely its signed weight by the update formula. If $k \notin J$, allowing k splits the new set into $\mathcal{A}_J$ and its translate by w_{k+1} exactly. The two pieces are disjoint, so adjoining one coordinate doubles the volume. At $J = \mathbb{N}$, the restricted set is the full set $\mathcal{A}$ by definition.

Theorem 7.2 (volume dichotomy for every allowed support)**.** *For every $J \subseteq \mathbb{N}$, exactly one of the following measure formulas applies:*

$$\begin{aligned} &J = F^c \text{ for a finite } F, && \lambda(\mathcal{A}_J) = 2^{-|F|}, \\ &J^c \text{ is infinite}, && \lambda(\mathcal{A}_J) = 0. \end{aligned}$$

For finite F, the division-free identity multiplies the face volume by $2^{|F|}$ and its solved form gives $2^{-|F|}$. Monotonicity under enlarging J is checked. It traps a set with infinitely many forbidden coordinates below finite-codimension faces of arbitrarily small dyadic measure, yielding measure zero. The two cases are assembled in the formal dichotomy. This theorem classifies the size of the value set generated inside any prescribed family of exponents. It does not classify the arithmetic nature of its individual points and therefore does not settle Problem 1.1.

7.2 The value $1/2$

A rational point of $\mathcal{A}$ attained by an infinite support refutes Problem 1.1. The distinguished candidate is $1/2$, and Theorem 7.3 gives a self-contained pair of exact alternatives.

The greedy expansion of $1/2$ is an exact finite computation as far as one cares to take it. Through level 21 it takes the exponents

$$2,\ 3,\ 6,\ 7,\ 14,\ 20,\ 21$$

and skips the others, the skipped runs being $\{1\}$, $\{4, 5\}$, $\{8, \dots, 13\}$ and $\{15, \dots, 19\}$; every level through 21 is survived. The second condition below asks whether the list of skipped exponents is infinite, and no finite computation answers that.

In the statement, $u = (u_1, \dots, u_d) \in \{0, 1\}^d$ is a finite prefix, $V(u) = \sum_{n \le d} u_n w_n$ is its value, and $T_n = \sum_{k > n} w_k$ as above.

Theorem 7.3 (membership and non-membership at $1/2$)**.** *The value $1/2$ belongs to $\mathcal{A}$ if and only if its canonical greedy expansion omits infinitely many exponents. Dually, $1/2 \notin \mathcal{A}$ is equivalent to the existence of a finite* fatal gap *— a prefix u through rank d with $V(u) + T_{d+1} < \frac{1}{2} < V(u) + w_{d+1}$, which makes every continuation miss — and to the greedy orbit having a last skipped exponent. Moreover no finite support has value $1/2$.*

Checked as the greedy form, the terminal-bit form, the skipped-rank form, the fatal-gap equivalence, its transfer to non-membership, and the finite-support exclusion.

Exact-row transport at the endpoint. The formal argument exposes the mechanism behind one conditional half-membership implication. Write r_{c-1} for the rational greedy remainder after ranks through $c - 1$. Whenever $c \ge 4$ and

$$0 < r_{c-1} < w_c,$$

the skipped-core constructor first turns the finite greedy core into an exact local quotient row at endpoint $2c - 2$, the local row constructor. The hard finite step is explicit: the deficit below $1/2$ is below the next Mersenne weight, so a Boolean word fills the entire upper-half capacity, the upper-half Boolean fill. The residual cannot vanish at any finite rank, the strict-positivity theorem; the odd denominator of every finite positive Mersenne sum rules out exact finite exhaustion of $1/2$. If positive skips occur cofinally in the exact sense the cofinal-skip hypothesis, their endpoints are cofinal by the row fan-in. The row-value estimate makes the discrepancy at endpoint n at most $(n + 1)/2^n$, and closedness then places $1/2$ in $\mathcal{A}$, the closed-set step.

No compatibility between the finite witnesses is assumed. The global forward endpoint theorem is in fact reversible.

Theorem 7.4 (positive skips characterise the half-value)**.** *The condition* `CofinalPositiveHalfGreedySkips` *holds if and only if $1/2 \in \mathcal{A}$.*

This is the checked equivalence. Thus cofinal positive skips are not an independently weaker hypothesis: proving them already proves the half-membership endpoint, while half-membership forces them because finite residuals never vanish. Neither side is established here, so the theorem identifies the exact obstruction without supplying an unconditional counterexample to Problem #257.

Theorem 7.5 (terminal scaled vanishing implies the half-value). *Let S be a* `HalfTerminalOnlyScaledVanishingSequence`*: its finite words exclude ranks* 0 *and* 1*, their depths tend to infinity, and the absolute terminal carry divided by* 2^M *tends to zero along the depths M. Then there is an infinite set $A \subseteq \mathbb{N}$ with*

$$\sum_{a\in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

Consequently the universal irrationality assertion in Problem 1.1 is false under this hypothesis. This is a conditional implication, not a construction of S and not a solution of Problem #257.

The terminal-only argument first places $1/2$ in the achievement set by the vanishing normalized-carry estimate, then uses the finite-support exclusion to show that the limiting support is infinite: achievement-set conclusion, infinite-support lift. The problem-level refutation is the direct formal consequence rational half counterexample, universal-claim refutation.

The formal source also proves equivalent terminal-bit and unbounded skipped-rank formulations in its internal half-cylinder seam coordinate; the two middle source links above are those versions. Their definitions are not needed for the self-contained greedy and fatal-gap statement used here. The two sides are asymmetric: non-membership is witnessed by a *finite* fatal gap and is therefore semi-decidable, whereas membership is an infinite condition. Computing further can only raise a lower bound on where a fatal gap could occur; it cannot establish survival.

The next local obstruction is sharper than a numerical search. Suppose a take occurs at rank b, the next take at rank $c > b+1$, and put $q = 2^b - 1$ and $m = 2^{c-1}$. If R is the reciprocal of the residual just before the first take, then the final intervening skip is dyadically unsafe exactly when

$$\frac{q(m-1)}{q+m-1} < R < \frac{qm}{q+m}.$$

The interval has the exact width $q^2/((q+m)(q+m-1))$. For a single intervening skip, $c = b+2$ and $m = 2q+2$, giving the two-thirds band

$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2},$$

whose width is below $1/9$. Thus the band is an exact localization of one greedy danger, not a claim that the actual orbit enters it.

Theorem 7.6 (the sharp local two-adic obstruction). *Let p, D, q be positive odd integers. If*

$$q(2q+1)p < 2D(3q+1), \qquad 2D(3q+2) < 2pq(q+1),$$

then $p \geq 7$. In particular, the odd numerator classes $p = 1, 3, 5$ cannot occupy the cleared two-thirds band. The bound is sharp for these local hypotheses: odd triples with $p = 7$ do occur in the band.

The hard step is genuinely 2-adic. The band defect $\Delta = 6D - 2pq$ is positive and divisible by 4 when p, D, q are odd, so $\Delta \geq 4$; combining that gap with the band inequalities forces $p > 6$. The exact formal boundaries are the general localization general band localization, the single-skip equivalence two-thirds band, and the sharp numerator conclusion odd numerator bound. Integral reciprocal states are excluded from the same band by integral safety. This is a one-step local no-go result: it neither proves that the half-greedy orbit reaches or avoids the band nor settles half-membership.

Two further conditional arguments sharpen the boundary around this half-value problem. They are logically distinct from the terminal scaled-vanishing question: one seeks irrationality from a structured support, while the other would produce a rational counterexample from cofinal geometric stages.

Theorem 7.7 (orthogonal-petal sunflower criterion). *Let $A \subseteq \mathbb{N}$ support an* `OrthogonalPetalBouquet` *and satisfy* `SunflowerForcedSlotTailSelection`*. Then*

$$\mathrm{Irrational}\left(\sum_{a \in A} \frac{1}{2^a - 1}\right).$$

The bouquet hypothesis packages a finite exceptional/core part and residual petals that are nontrivial, pairwise coprime, and reciprocal-summable. The remaining selector hypothesis is a uniform tail condition: at every positive length K it supplies a shifted block whose binary carry is divisible by 2^K while its residual tail is bounded. Neither hypothesis is constructed here, so this is a conditional reduction rather than a solution of Problem #257.

The hard mechanism is the persistent-reduced-modulus averaging argument: the selector supplies carries, and the support-series theorem turns that supply into irrationality. The exact formal consumers are forced carry supply and irrationality endpoint. The source definition of the selector makes the missing analytic obligation explicit at uniform tail selector.

Theorem 7.8 (the composite-dilation defect). *Let $A \subseteq \mathbb{N}$ and let $a, x \in \mathbb{N}_{>0}$ with $a \in A$. Define the foreign-divisor defect*

$$\delta_A(a, x) = \#\{d \mid ax : d \in A,\ d \nmid x,\ d \neq a\}.$$

Then the divisor incidence satisfies the exact identity

$$\mathrm{sc}_A(ax) = \mathrm{sc}_A(x) + \mathbf{1}_{a \nmid x} + \delta_A(a, x).$$

If every member of A is prime, then $\delta_A(a, x) = 0$, so the familiar prime-support dilation formula is recovered. More generally, if A has an `OrthogonalPetalBouquet` *witness h_B, then for every $i \in \mathbb{N}$,*

$$\delta_A(\mathrm{ray}_{h_B}(i), x) \leq |h_B.\mathrm{exceptional}| + \mathrm{sc}_{\mathrm{range}(h_B.\mathrm{petal})}(x),$$

where $\mathrm{ray}_{h_B}(i) = h_B.\mathrm{core}(i) h_B.\mathrm{petal}(i)$.

The first identity is the finite divisor partition checked at exact dilation identity. The hard point is that a composite multiplier can create support divisors which are neither the multiplier itself nor old divisors of x; they are recorded by δ_A rather than silently discarded. The defect vanishes under the prime-support hypothesis by prime-support no-defect, and the corresponding incidence formula is prime specialization. For a bouquet, every non-exceptional foreign divisor injects into a unique petal divisor of x, yielding the displayed budget through foreign-divisor classification and defect bound. The correction is concrete already for $A = \{2, 6\}$: multiplying $x = 1$ by the composite support element 6 creates the additional support divisor 2, so $\delta_A(6, 1) = 1$ (two-six witness).

Thus a composite bouquet cannot be analysed by copying the prime formula until this explicit foreign channel has been budgeted. This theorem provides that local budget only; it neither bounds defects along an infinite orbit nor supplies the uniform tail selector required by Theorem 7.7.

Theorem 7.9 (cofinal cylinders imply an infinite half-support)**.** *Suppose that for every N there are M, K with* $\max\{N, 1\} \leq M$ *and a nonempty* `CylinderStage` *K M. Then there is an infinite set $A \subseteq \mathbb{N}$ with $0 \notin A$ and*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

Thus the universal irrationality assertion in Problem #257 is false under this cofinal-stage hypothesis. The formal proof does not construct the cofinal family of full-cylinder stages.

The proof first converts each full-cylinder stage into a weaker terminal-only strip, then invokes closedness of the half-achievement set and the positive-support lift. The exact steps are terminal-only bridge, infinite half-support, and positive-support lift. This is a conditional implication, not evidence that such stages exist; proving or refuting their cofinality is the remaining task for this argument.

One natural construction for $1/2$ is ruled out. The Boolean support selected by the negative values of the Möbius function has value exactly $1/2$ plus the positive Möbius tail, hence at least $1/2 + 1/63$ (identity, bound). That closes the sign-truncation construction; it excludes no other support.

7.3 A second rational target

The target $1/21$ has a useful property that does not depend on any finite search.

Theorem 7.10 (finite-support obstruction at $1/21$)**.** *For every finite set F of exponents with $n \geq 2$ for all $n \in F$,*

$$\sum_{n \in F} \frac{1}{2^n - 1} \neq \frac{1}{21}.$$

This is checked by the formal theorem. The reduced denominator 21 has doubling order 6, so the exact denominator–lcm identity forces every selected exponent to divide 6. The lower bound $n \geq 2$ leaves only $2, 3, 6$, and the remaining eight subsets are discharged by exact rational arithmetic. Thus any representation of $1/21$ along this rank-≥ 2 route, if one exists, must be infinite. The theorem proves nontermination only: it does not prove that $1/21$ lies in $\mathbb{A}$.

A separate arithmetic lemma isolates the primitive cone that appears in one candidate expansion. Write

$$\mathcal{P}_{23}(n) = \{(p,q) \in \mathbb{N}^2_{>0} : \gcd(p,q) = 1,\ 2p + 3q = n\}.$$

Every $n \geq 11$ has a member of $\mathcal{P}_{23}(n)$ (checked), whereas $\mathcal{P}_{23}(10)$ is empty (checked). Multiplicity begins immediately at 11, and every $10k$ with $k \geq 2$ has at least two distinct primitive representations (rank eleven, multiples of ten). These are exact Diophantine facts, not a counterexample construction. In particular the recurring collisions show why primitive-cone coverage is not already a Boolean expansion: the missing step is a proof that the relevant integer multiplicities can be converted into zero–one reciprocal-Mersenne digits without changing the value.

7.4 The scaled-greedy criterion at $1/21$

The finite-support obstruction makes $1/21$ useful only if the infinite problem is stated in coordinates that retain the actual greedy orbit. Let

$$r_N = \mathrm{greedyMersenneRemainder}(1/21, N)$$

be the real remainder after rank N, put $y_N = 2^N r_N$, and set

$$c_{N+1} = \frac{2^{N+1}}{2^{N+1} - 1}.$$

The scaled recurrence is a nearly doubling map: it sends y_N to $2y_N - c_{N+1}$ when $c_{N+1} \leq 2y_N$, and to $2y_N$ otherwise. The latter is the exact lower branch, so rank $N+1$ is skipped precisely when $2y_N < c_{N+1}$.

Theorem 7.11 (scaled-greedy trap, escape, and cofinal return)**.** *For every nonnegative real x, writing $y_N(x) = 2^N$ greedyMersenneRemainder(x, N),*

$$x \in \mathcal{A} \iff y_N(x) < 2 \quad \textit{for every } N.$$

If $x \notin \mathcal{A}$, then $y_N(x) \to +\infty$. Conversely, a single bounded cofinal subsequence already characterizes membership:

$$x \in \mathcal{A} \iff \exists B \in \mathbb{R}\ \forall K\ \exists N \geq K : \qquad y_N(x) \leq B.$$

For every nonnegative rational q this is also equivalent to

$$\forall K\ \exists N \geq K : \qquad 2y_N(q) < \frac{2^{N+1}}{2^{N+1} - 1}.$$

In particular, $1/21 \in \mathcal{A}$ if and only if its scaled greedy orbit crosses this moving lower separatrix beyond every cutoff.

The trap-set equality is kernel checked, the escape theorem is kernel checked, and the bounded-cofinal-return equivalence is kernel checked. The escape proof locates a fatal rank at which the residual exceeds the complete remaining tail by some $\delta > 0$. That excess persists, so the scaled residual is at least $2^N\delta$ thereafter. This is the hard step in the reverse bounded-return implication: an orbit with even one bounded cofinal subsequence cannot lie outside $\mathcal{A}$.

The general rational criterion is kernel checked, and its $1/21$ specialization is kernel checked. The hard direction uses the rational normal form: membership is equivalent to infinitely many omitted exponents, and a skip is exactly a lower-separatrix crossing. This is an exact reformulation, not a proof that the crossings occur. It asks for neither convergence nor a prescribed uniform bound.

The quotient coordinate gives a complementary classification of what nonmembership would have to look like. Let

$$Q_N = \left\lfloor \frac{2^N}{21} \right\rfloor - P_N,$$

where P_N is the integral numerator of the binary divisor-incidence prefix generated by that same greedy support. Thus Q_N is the nonnegative denominator-21 defect after the six-periodic residue of 2^N mod 21 has been removed. These are not independent models: the formal source proves an exact identity relating Q_N, $2^N r_N$ and a finite-prefix divisor tail (checked).

A finer denominator-specific coordinate works directly with the denominator-21 quotient-greedy orbit. At even depth $2R$, write

$$\begin{aligned} D_R &= \text{twentyOneEvenQuotientGreedySupport}(R), \\ s_R &= \text{twentyOneEvenQuotientGreedyRemainder}(R). \end{aligned}$$

Here $D_R \subseteq \{2, \dots, R\}$ is the Boolean support obtained by descending integer-greedy selection on the exact scaled quotient weights, and s_R is its terminal scalar remainder (support, remainder). Write $\mathcal{F}_{21}$ for `TwentyOneFatalAlignedBranch`, the explicit branch in which the real greedy orbit has a fatal witness, only finitely many skipped exponents (hence cofinite eventual selection), eventual quotient/rational-greedy alignment, and eventual occupation of every doubling block.

Theorem 7.12 (fatal-branch quotient refinement at $1/21$)**.** *The following statements hold.*

1. $1/21 \in \mathcal{A}$ *if and only if* $\mathcal{F}_{21}$ *does not hold.*
2. *If there is an unbounded sequence of ranks R with $s_R \le 2^R$, then* $1/21 \in \mathcal{A}$.
3. *On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R+1$ belongs to D_{R+1}, and (D_R, s_R) follows one exact affine recurrence.*

The equivalence is kernel checked; the closed-row compactness step is kernel checked; and the eventual affine regime is kernel checked. Explicitly, on $\mathcal{F}_{21}$ the quotient orbit is eventually strictly supercapacity and loses its last Boolean choice: if τ_R is the periodic target pulse and π_R the divisor pulse generated by D_R, then for every sufficiently large R,

$$s_R > 2^R, \qquad D_{R+1} = D_R \cup \{R+1\}, \qquad s_{R+1} = 4s_R + \tau_R - \pi_R - (2^{R+1} + 1).$$

The denominator-specific separation theorem additionally proves that every closed Boolean quotient row is exactly the canonical quotient-greedy row, including exact saturation at the boundary (kernel checked). An aligned crossing from saturation into strict supercapacity forces a missing canonical ancestor at one-third or two-thirds scale and a real skipped exponent with square-root-bounded defect (ancestor hole, scaled skip).

These conclusions remove alternative late Boolean branches; they do not exclude the full fatal/cofinite/aligned branch. On that branch the permanent affine-supercapacity recurrence is forced, so a contradiction of the recurrence would be sufficient, but the recurrence alone is not the exact membership endpoint. It is conditional, not a contradiction: synthetic pulses can sustain such an affine recurrence without being the actual divisor pulse.

8 Open problems

Problem 1.1 remains the frame: the settled families in Section 5 do not approach a universal quantifier, and the forced conditions in Section 4 are not known to be contradictory. Theorem 1.2 does, however, force every counterexample into the reciprocal-divergent region $\sum_{a\in A} a^{-1} = \infty$. For the base-2 universal problem, three endpoint questions organise the remaining discussion. The universal problem and the half-value question are not independent, and the relation between them is asymmetric. No finite support has value $1/2$ (Theorem 7.3), so $1/2 \in \mathcal{A}$ would exhibit an *infinite* support with a rational value and thereby refute Problem 1.1; equivalently, a positive answer to Problem 1.1 puts $1/2$ outside $\mathcal{A}$. The converse direction does not follow: $1/2 \notin \mathcal{A}$ would give a finite fatal-gap witness and eliminate this candidate, but would not imply the universal statement. The half-value question is therefore a one-sided test of #257, not a second problem beside it.

Problem 8.1 (membership of 1/21 in the Mersenne achievement set)**.** Prove cofinal crossings of the exact moving lower separatrix

$$2\,\mathrm{scaledGreedyRemainder}(1/21, N) < \mathrm{mersenneScale}(N+1).$$

Equivalently, exclude the explicit fatal/cofinite/aligned branch $\mathcal{F}_{21}$. It is sufficient either to contradict the eventual permanent affine-supercapacity recurrence forced by that branch or to force an unbounded sequence of closed canonical quotient rows; neither sufficient route is claimed to be equivalent by itself.

1. **Problem 1.1 itself**, for arbitrary infinite A. Theorem 1.2 excludes every reciprocal-summable support, so an unresolved counterexample must have divergent reciprocal mass. Section 4 constrains every hypothetical counterexample without excluding one in that remaining region. Every counterexample would have unbounded scaled-tail states, sublogarithmic divisor-coverage gaps and an admissible Boolean–Möbius scaled-tail sequence. The conditional lower bounds in Theorem 4.3 are now subordinate inside the proved reciprocal-summable exclusion rather than the frontier itself. No contradiction is known in the reciprocal-divergent region.

2. **Membership of $1/2$ in $\mathcal{A}$.** Theorem 7.3 makes this exactly equivalent to infinitely many greedy skips, and to a fatal gap on the other side; neither is proved. A proof of non-membership would take the form of one finite fatal gap.

3. **Problem 8.1.** Theorem 7.10 rules out finite support, while Theorem 7.12 reduces membership exactly to excluding $\mathcal{F}_{21}$. Contradicting its forced eventual affine-supercapacity recurrence or producing arbitrarily deep closed canonical rows would suffice. Neither input is known, and neither is promoted to an equivalence.

The half-value question remains valid on both sides; the criterion at $1/21$ is at present the sharper of the two. The questions below refine that criterion and the arithmetic constraints around it, in order of how directly an answer would move the proved boundary.

Problem 8.2 (permanent affine supercapacity)**.** Can the *actual* denominator-21 greedy orbit satisfy $\mathcal{F}_{21}$ indefinitely? Equivalently, can the complete fatal, cofinite-selection, alignment and doubling-block hypotheses coexist with the eventual recurrence

$$s_R > 2^R, \quad D_{R+1} = D_R \cup \{R+1\}, \quad s_{R+1} = 4s_R + \tau_R - \pi_R - (2^{R+1} + 1),$$

or must an arbitrarily deep closed return $s_R \le 2^R$ occur?

A contradiction from a Lyapunov function, a 2-adic obstruction, a divisibility theorem or a recurrence classification would prove $1/21 \in \mathcal{A}$ and, by Theorem 7.10, produce an infinite rational support. Conversely, an actual construction satisfying *all* clauses of $\mathcal{F}_{21}$ would prove $1/21 \notin \mathcal{A}$. Constructing only an abstract affine orbit with adversarial pulses does neither.

Problem 8.3 (weakest native recurrence criterion)**.** Does the scaled actual greedy remainder return cofinally to one bounded interval?

$$\exists B < \infty \ \forall K \ \exists N \ge K : \qquad 2^N r_N \le B.$$

This condition is both necessary and sufficient. It is the $x = 1/21$ specialization of the general membership equivalence in Theorem 7.11 (checked). It asks for neither convergence, a prescribed B, a global bound nor bounded return gaps. Two concrete stronger targets are cofinal recurrence of $Q_N \le 1$ (sufficient condition) and arbitrarily deep closed quotient rows $s_R \le 2^R$. Exact computation through rank 200,000 records 96 zero-defect returns and 4,956 returns to $Q_N \le 1$, with last observed ranks 193,690 and 199,930 and maximum observed gap 492; these finite data do not prove cofinality.

Problem 8.4 (actual-orbit invariant)**.** Is there a finite-memory, 2-adic or discrepancy invariant, using the correlated divisor pulses of the actual support, that forces a closed return or forbids permanent supercapacity? More precisely, can one use a bounded window of R mod 6, residues of s_R, endpoint divisor counts and the finite set of eventual skips to force descent or a forbidden state? Alternatively, can one prove that no bounded-memory invariant distinguishes the true orbit from synthetic permanent-supercapacity controls?

Pointwise pulse bounds are known to be too weak. The exact six-step recurrence is

$$Q_{N+6} = 64Q_N + 3(2^N \bmod 21) - L_N,$$

with L_N the actual weighted repair load (checked). The formerly plausible translation-invariant six-step contraction first fails at $N = 73$; the surviving statement is the slope-aware repair-load iff at the exact equivalence.

Problem 8.5 (final-skip Diophantine exclusion)**.** Let $E = \sum_{n\geq 1}(2^n - 1)^{-1}$. If $1/21 \notin \mathcal{A}$, let M be the last skipped exponent, S_M its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d\in S_M} \frac{1}{2^d - 1}.$$

Can the complete final-skip signatures be used to prove $|E - a_M| \geq \mathrm{gap}_M$, contradicting

$$0 < a_M - E < \mathrm{gap}_M\,?$$

The fatal interval is checked as an exact one-sided approximation. The reduced skipped-prefix denominator has binary order equal to the lcm of the skipped exponents and hence at least M (order identity, lower bound). Parity-sensitive gcd restrictions and adjacent-numerator product inequalities are also checked, but they are necessary signatures rather than the missing upper-height or irrationality-measure estimate.

Problem 8.6 (classification of rational targets)**.** For $L \geq 1$, classify the reduced rationals

$$\mathcal{T}_L = \left\{ \frac{p}{q} \in (0,1) : \begin{array}{l} \gcd(p,q) = 1,\ q \text{ odd},\ \mathrm{ord}_q(2) \leq L, \\ p/q \text{ has no finite Mersenne representation} \end{array} \right\}$$

for which non-membership reduces to an eventually deterministic finite-memory or affine quotient recurrence. In particular, classify $\mathcal{T}_6$ by target-pulse alphabet, finite-support obstruction, weakest membership criterion, surviving branch complexity and available Diophantine estimate. Is $1/21$ minimal or unique under explicit criteria, or does another target have a strictly simpler fatal branch?

Problem 8.7 (realised finite denominators)**.** For fixed $b \geq 2$ and $L \geq 2$, let $\mathcal{D}_b(L)$ be the reduced denominators of the nonempty finite sums $\sum_{n\in F}(b^n - 1)^{-1}$ with $\mathrm{lcm}(F) = L$. Is

$$\mathcal{D}_b(L) = \{D : D \mid b^L - 1,\ \mathrm{ord}_D(b) = L\}?$$

If not, what cyclotomic, valuation or cancellation conditions characterise the realised denominators?

The order theorem supplies the inclusion from left to right, not its converse. A counterexample and corrected classification, or effective extremal bounds within $\mathcal{D}_b(L)$, would strengthen the lead theorem and feed height information back into Problem 8.5. For scope, all squarefree values at power-of-two bases, jointly in each finite family, are settled by Corollary 1.2 and Example 1.1 of [4, author-preprint p. 4] (Corollary 6.2). Their cited corollary does not cover bases that are not powers of 2, and this note makes no global open-status or priority claim for those values.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions; kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. The displayed proof of Corollary 6.3, the derivation of Corollary 6.2 from [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4], and the general finite-change argument (from the two checked prefix lemmas) are expository arguments rather than named checked statements.

The squarefree no-go interfaces, shifted coefficient, shift equivalence, Chinese-remainder block supply, and restricted-selector injectivity are directly checked statements linked at their use; they are not prose-only claims.

The numerical instances — the table of finite supports in Section 3, the incidence and zero-window examples, the reciprocal-mass and Boolean–Möbius instances, the 3/4 certificate, and the greedy expansion of 1/2 through level 21 — are direct computations, reported as computations and not as theorems. Results attributed to Campbell, to Duverney and Tachiya, to Tao and Teräväinen, to Luca and Tachiya, and to Kovač and Tao are cited from the literature and are not formalised here.

Erdős Problem #257 remains open.

References

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[2] D. Duverney, Y. Suzuki and Y. Tachiya, *Linear independence results for certain sums of reciprocals of Fibonacci and Lucas numbers*, arXiv:1908.07290, 2019. Corollary 1 gives the stated sparse reciprocal-sum linear independence under the pairwise-coprime, odd-support and reciprocal-summability hypotheses.

[3] J. M. Campbell, *On the binary digits of the Erdős–Borwein constant*, arXiv:2605.24160v1, 2026. Theorem 1 on p. 12 proves that the binary block 11 occurs infinitely often in the base-2 expansion of the full-support value; its proof occupies pp. 12–24.

[4] D. Duverney and Y. Tachiya, *Refinement of the Chowla–Erdős method and linear independence of certain Lambert series*, Forum Math. 31 (2019), no. 6, 1557–1566, DOI. Corollary 1.2 gives the general $F_s(E)$ theorem and its monomial images under $|q|^{\mathrm{lcm}(1,\ldots,\ell)} \leq s$; Example 1.1 gives the joint squarefree family at all bases 2^j, $j \geq 1$. Both are on p. 4 of the linked author preprint; the proof is on pp. 9–11.

[5] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Theorem 1.3 proves $\sum_{n\geq 1} \omega(n)/2^n = \sum_p (2^p - 1)^{-1}$ irrational, settling the prime-support case of #257 at base 2; the extension to every integer base and the prime-power support are stated as remarks with the modifications left to the reader. The theorem is on p. 4 and its proof is Section 5, pp. 44–56, in arXiv v2.

[6] F. Luca and Y. Tachiya, *Linear independence results for the values of divisor functions series*, RIMS Kôkyûroku No. 2014 (2017), 138–150. Theorem A on p. 139 explicitly restates their earlier Theorem 1.1 for nonzero purely periodic integer weights. Theorem 1 is on p. 139, its examples are on p. 140, and its proof is on pp. 149–150.

[7] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. 175 (2025), 572–608, DOI. Remark 4.1 (p. 13) records the strict tail inequality and the Cantor structure. Theorem 2.3 (p. 5; proof pp. 13–14) constructs rational merged sums from several bases under its mass hypothesis; it is not a fixed-base counterexample.

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[12] T. F. Bloom, *Erdős Problem #257*, `erdosproblems.com/257`, accessed 28 July 2026 (page displays "last edited 15 April 2026"). The current record labels the universal fixed-base problem open, cites `[Er68d]`, `[ErGr80, p. 62]` and `[Er88c, p. 105]`, records the Tao–Teräväinen prime and prime-power cases and the Kovač–Tao refutation of a broader varying-denominator speculation, and explicitly describes its status as the website owner's present assessment rather than a literature-completeness guarantee.